

Maharashtra Public Service Commission Assessment

Question Paper Name :	Assistant And Associate Professor Civil Engineering Maharashtra Engg Teachers Ser G A 27Apr26 S1
Subject Name :	Assistant And Associate Professor Civil Engineering Maharashtra Engg Teachers Service Group A
Creation Date :	2026-04-27 14:33:09
Duration :	60
Number of Questions :	100
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes

Assistant And Associate Professor Civil Engineering Maharashtra Engg Teachers Service Group A

Group Number :	1
Group Id :	15056117
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0

Assistant And Associate Professor Civil Engineering Maharashtra Engg Teachers Service Group A

Section Id :	15056120
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100
Number of Questions to be attempted :	100
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	15056125
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 1505611603 Question Type : MCQ Option Shuffling : No

The value of $\nabla^2 (r^n)$ is _____.

Options :

1. $n(n+1)r^{n-2}$
2. $n(n-1)r^{n-1}$
3. nr^{n-2}
4. $(n+1)r^{n-2}$

Question Number : 2 Question Id : 1505611604 Question Type : MCQ Option Shuffling : No

For a normal distribution with mean 50 and standard deviation 10, the value $x = 70$ corresponds to a Z – score of _____.

Options :

1. 1
2. 2
3. 3
4. - 2

Question Number : 3 Question Id : 1505611605 Question Type : MCQ Option Shuffling : No

The workdone by a force field along different paths between two points is path independent if the field is _____.

Options :

1. Solenoidal
2. Conservative
3. Rotational
4. Divergent

Question Number : 4 Question Id : 1505611606 Question Type : MCQ Option Shuffling : No

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$, then the rank of matrix A is _____.

Options :

1. 0
2. 1
3. 2
4. - 1

Question Number : 5 Question Id : 1505611607 Question Type : MCQ Option Shuffling : No

The integrating factor of $\frac{dy}{dx} + 2y = e^x$ is _____.

Options :

1. e^{-2x}
2. e^{-x}
3. e^{2x}
4. x^2

Question Number : 6 Question Id : 1505611608 Question Type : MCQ Option Shuffling : No

The probability density function of a normal distribution is symmetric about _____.

Options :

1. Zero
2. Mean
3. Variance
4. Mode

Question Number : 7 Question Id : 1505611609 Question Type : MCQ Option Shuffling : No

Simpson's $\frac{1}{3}$ rule uses _____.

Options :

1. Linear interpolation
2. Cubic interpolation
3. Quadratic interpolation
4. Exponential interpolation

Question Number : 8 Question Id : 1505611610 Question Type : MCQ Option Shuffling : No

Let the probability that an individual suffers a bad reaction from an injection is 0.001. What is probability that out of 3000 individuals exactly 3 individuals will suffer a bad reaction ?

Options :

1. $4.5e^3$
2. $4.5e^{-3}$
3. $5.4e^3$
4. $5.4e^{-3}$

What is the type of partial differential equation

$$y^2 u_{xx} - 2xy u_{xy} + x^2 u_{yy} = \frac{y^2}{x} u_x + \frac{x^2}{y} u_y ?$$

Options :

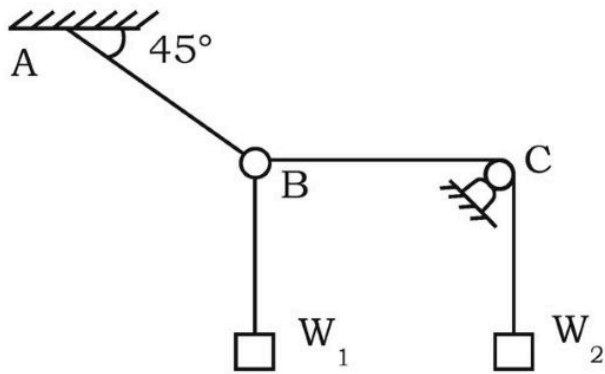
1. Hyperbolic
2. Parabolic
3. Elliptic
4. None of the above

The differential equation $(y^2 e^{xy^2} + 6x) dx + (2xye^{xy^2} - 4y) dy = 0$ is _____.

Options :

1. Linear, homogeneous and exact
2. Non-linear, homogeneous and exact
3. Non-linear, non-homogeneous and exact
4. Non-linear, non-homogeneous and non-exact

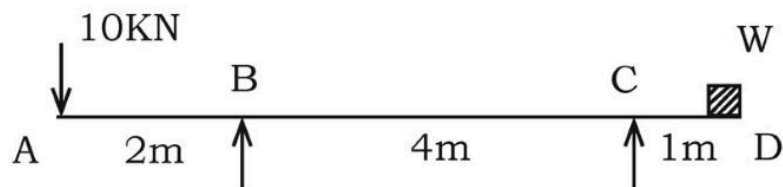
What shall be the relation between W_1 and W_2 , for the following cable system to be in equilibrium ? 'B' is a ring and 'C' is a frictionless pulley.



Options :

1. $W_1 = W_2$
2. $W_1 = W_2 \cdot \sqrt{2}$
3. $W_1 = \frac{W_2}{\sqrt{2}}$
4. $W_1 = 2W_2$

B and C are knife edge supports. What must be range of 'W' for equilibrium ?



Options :

1. $10\text{kN} \leq W \leq 40 \text{ kN}$
2. $5 \text{ kN} \leq W \leq 12 \text{ kN}$
3. $4 \text{ kN} \leq W \leq 20 \text{ kN}$
4. $4 \text{ kN} \leq W \leq 60 \text{ kN}$

Question Number : 13 Question Id : 1505611615 Question Type : MCQ Option Shuffling : No

The translatory effect of the couple is _____.

Options :

1. Maximum
2. Zero
3. Minimum
4. Average

Question Number : 14 Question Id : 1505611616 Question Type : MCQ Option Shuffling : No

Following is not affecting bond resistance in RCC structures.

Options :

1. Grade of concrete
2. Diameter of bar
3. Bar profile condition
4. Spacing of reinforcement

Question Number : 15 Question Id : 1505611617 Question Type : MCQ Option Shuffling : No

A column has actual length of 6 m. It's both ends are fixed. What is it's equivalent length based on adjacent points of inflexion ?

Options :

1. 3 m
2. 6 m
3. $6\sqrt{2}$ m
4. None of the above

Question Number : 16 Question Id : 1505611618 Question Type : MCQ Option Shuffling : No

Shear modulus is the ratio of _____.

Options :

1. Linear stress to linear strain
2. Shear stress to shear strain
3. Linear stress to lateral strain
4. Volumetric strain to linear stress

Question Number : 17 Question Id : 1505611619 Question Type : MCQ Option Shuffling : No

The systematic development of consistent deformation method in the matrix form has lead to _____.

Options :

1. Flexibility method
2. Stiffness method
3. Displacement method
4. Equilibrium method

Question Number : 18 Question Id : 1505611620 Question Type : MCQ Option Shuffling : No

In case of moment distribution method the ratio of carry over moment to applied moment is called as _____.

Options :

1. Distribution factor
2. Carry over factor
3. Stiffness factor
4. Flexibility factor

Question Number : 19 Question Id : 1505611621 Question Type : MCQ Option Shuffling : No

Which of the following is/are included in the data required for mix proportioning ?

- a. Grade of concrete
- b. Maximum size of aggregates
- c. Minimum cement content
- d. Maximum cement content

Options :

1. Only a and c
2. Only a and b
3. Only a, b and c
4. All of the above

In moment distribution method, the sum of distribution factor of all the members meeting at any joint is always _____.

Options :

1. Zero
2. < 1
3. > 1
4. $= 1$

Which of the following are considered as same for Fe 250 and Fe 415 ?

- a. Ductility
- b. Modulus of Elasticity
- c. Poisson's ratio
- d. Strength
- e. Carbon content
- f. Density
- g. Co-efficient of thermal expansion

Options :

1. a, d and e
2. a, b, c, d, e, f and g
3. b, c, f and g
4. c, e and g

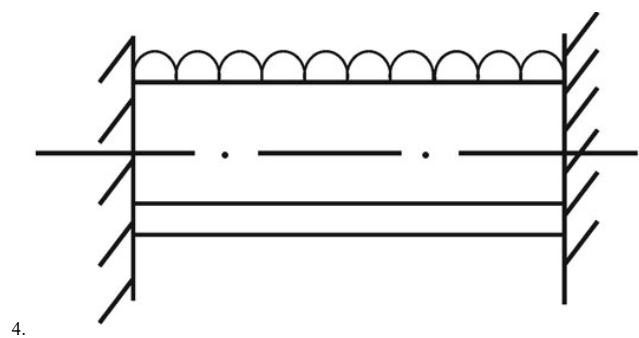
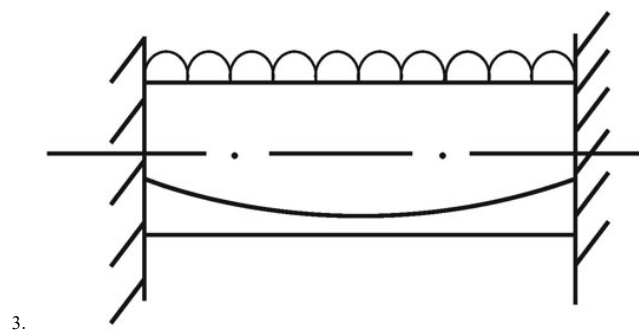
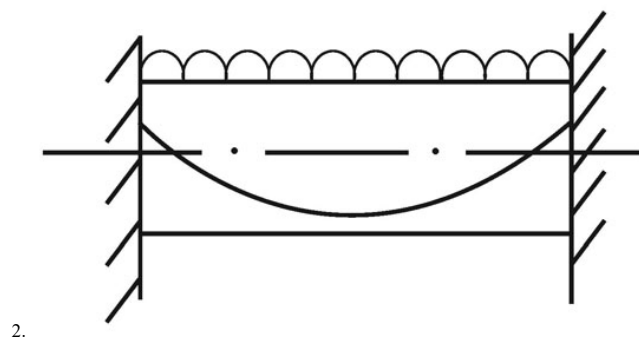
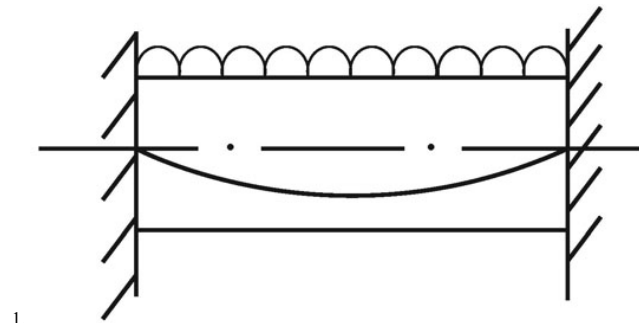
For M49 grade of concrete the short term and long term modulus of elasticity of concrete will be _____ respectively. (Take creep co-efficient = 1).

Options :

1. 49 MPa, 35 GPa
2. 35 GPa, 17.5 GPa
3. 7 MPa, 49 MPa
4. 21 GPa, 10.5 GPa

A prestressed concrete beam, fixed at ends, is subject to uniformly distributed load. Which of the following is most suitable load balancing cable ?

Options :



Question Number : 24 Question Id : 1505611626 Question Type : MCQ Option Shuffling : No

In case of laterally supported steel beams, the permissible bending stress in tension (σ_{bt}) and maximum permissible bending stress in compression (σ_{bc}) should not exceed _____.

Options :

1. 0.55 f_y
2. 0.66 f_y
3. 0.77 f_y
4. 0.88 f_y

Question Number : 25 Question Id : 1505611627 Question Type : MCQ Option Shuffling : No

What is/are factor/s affecting design shear strength of R.C.C. beam ?

- a. Cross-sectional size
- b. Area of tension reinforcement
- c. Grade of concrete
- d. Span of the beam

Options :

1. All of a, b, c and d
2. Only a, b and c
3. Only a and c
4. Only a and d

Question Number : 26 Question Id : 1505611628 Question Type : MCQ Option Shuffling : No

The water content corresponding to maximum density is called as _____.

Options :

1. Maximum water content
2. Optimum water content
3. Zero water content
4. Saturated water content

Question Number : 27 Question Id : 1505611629 Question Type : MCQ Option Shuffling : No

The most accurate method for the determination of water content in laboratory is _____.

Options :

1. Sand Bath Method
2. Oven Drying Method
3. Pycnometer Method
4. Calcium Carbide Method

Question Number : 28 Question Id : 1505611630 Question Type : MCQ Option Shuffling : No

Terzaghi demonstrated the spring analogy theory to understand the mechanics of _____.

Options :

1. Compaction
2. Seepage
3. Consolidation
4. Permeability

Question Number : 29 Question Id : 1505611631 Question Type : MCQ Option Shuffling : No

Poisson's ratio of soil sample is 0.4, using theory of elasticity, estimated value of coefficient of lateral earth pressure at rest in same soil is _____.

Options :

1. 0.55
2. 0.66
3. 0.33
4. 1.0

Question Number : 30 Question Id : 1505611632 Question Type : MCQ Option Shuffling : No

When allowable soil pressure is low or building load are heavy, suitable type of foundation is _____.

Options :

1. Strap footing
2. Raft footing
3. Spread footing
4. Combined footing

Question Number : 31 Question Id : 1505611633 Question Type : MCQ Option Shuffling : No

Correct formula for calculation of saturated unit weight of soil mass is, where, G = specific gravity of soil

e = void ratio

γ_w = unit weight of water

Options :

1.
$$\gamma_{\text{sat}} = \left[\frac{G - e}{1 + e} \right] \gamma_w$$

2.
$$\gamma_{\text{sat}} = \left[\frac{G + e}{1 + e} \right] \gamma_w$$

3.
$$\gamma_{\text{sat}} = \left[\frac{G + e^2}{1 + e^2} \right] \gamma_w$$

4.
$$\gamma_{\text{sat}} = \left[\frac{G + 1}{1 + e} \right] \gamma_w$$

Question Number : 32 Question Id : 1505611634 Question Type : MCQ Option Shuffling : No

Sample produced in sandy soil by split-spoon sampler is _____.

Options :

1. Disturbed sample

2. Undisturbed sample

3. Representative sample

4. Test sample

Question Number : 33 Question Id : 1505611635 Question Type : MCQ Option Shuffling : No

A shallow foundation is usually defined as foundation which has _____.

Options :

1. Depth less than 0.6 m

2. Depth less than 1 m

3. Depth less than or equal to width

4. Depth less than 1.2 m

Question Number : 34 Question Id : 1505611636 Question Type : MCQ Option Shuffling : No

For a same soil, the effect of heavier compaction is to increase maximum dry density and _____.

Options :

1. increase optimum water content
2. decrease optimum water content
3. does not affect optimum water content
4. decreases more density

Question Number : 35 Question Id : 1505611637 Question Type : MCQ Option Shuffling : No

For reducing the driving force, the outside clearance of sampling tube lies between _____ and _____ percent.

Options :

1. 0, 1
2. 0.5, 1.5
3. 0, 2
4. 0, 3

Question Number : 36 Question Id : 1505611638 Question Type : MCQ Option Shuffling : No

Bishop's method of stability analysis assumes _____.

Options :

1. The slip surface is an arc of circle
2. Soil is stratified
3. The forces acting on the sides of slices is neglected
4. All of the above

Question Number : 37 Question Id : 1505611639 Question Type : MCQ Option Shuffling : No

The ratio of the volume of voids to the total volume of soil is _____.

Options :

1. Voids ratio
2. Degree of saturation
3. Air content
4. Porosity

The maximum particle size for which Darcy's law is applicable is _____.

Options :

1. 0.2 mm
2. 0.1 mm
3. 0.5 mm
4. 2.0 mm

The coefficient of compressibility is the ratio of _____.

Options :

1. Stress to strain
2. Decrease in voids ratio to increase in effective stress
3. Volumetric stress to volumetric strain
4. Change in thickness to change in stress

A pile foundation is used when _____.

Options :

1. the loads are heavy
2. the soil stratum near ground surface is weak
3. both (1) and (2)
4. neither (1) nor (2)

Furrow irrigation is widely used for _____ crops.

Options :

1. Row
2. Column
3. Seasonal
4. All of the above

Question Number : 42 Question Id : 1505611644 Question Type : MCQ Option Shuffling : No

Gravity dams are in which the water pressures are resisted by _____.

Options :

1. Weight of the dam and wind pressure
2. Weight of the dam only
3. Water pressure from down stream and weight of the dam
4. Weight of the dam, wind pressure and water pressure from down stream

Question Number : 43 Question Id : 1505611645 Question Type : MCQ Option Shuffling : No

The topography in which there is non availability of space, the type of spillway preferred is _____.

Options :

1. Straight drop spillway
2. Chute Spillway
3. Shaft Spillway
4. Ogee Spillway

Question Number : 44 Question Id : 1505611646 Question Type : MCQ Option Shuffling : No

Which irrigation method is useful for sandy soil, shallow land and where leveling is impossible ?

Options :

1. Subsurface Irrigation
2. Drip Irrigation
3. Sprinkler Irrigation
4. Surface Irrigation

Pitot static tube is used to measure _____ .

Options :

1. Velocity of flow
2. Volume flow rate
3. Humidity of air
4. Temperature of fluid

In case of Laminar flow _____ .

Options :

1. Forces due to viscosity predominates over the inertial forces
2. Forces due to inertial forces predominates the viscous forces
3. Viscous forces and inertial forces are equal
4. None of the above

Froude Number is given by _____ .

Options :

1. $\frac{VL}{V}$
2. $\frac{V}{\sqrt{p/Q}}$
3. $\frac{V}{\sqrt{k/Q}}$
4. $\frac{V}{\sqrt{g.L}}$

Question Number : 48 Question Id : 1505611650 Question Type : MCQ Option Shuffling : No

For $Fr_1 = 2.5$ to 4.5 , the hydraulic jump formed is known as _____.

Options :

1. an undular jump
2. weak jump
3. steady jump
4. an oscillating jump

Question Number : 49 Question Id : 1505611651 Question Type : MCQ Option Shuffling : No

The boundary of discontinuity between the two air masses is termed as _____.

Options :

1. Barriers
2. Front
3. Diurnal variation
4. None of the above

Question Number : 50 Question Id : 1505611652 Question Type : MCQ Option Shuffling : No

The effective rainfall is also termed as _____.

Options :

1. Rainfall deficiency
2. Rainfall excess
3. Rainfall pattern
4. Rainfall distribution

Question Number : 51 Question Id : 1505611653 Question Type : MCQ Option Shuffling : No

The volume of water released or taken into storage by it per unit surface area of the aquifer per unit decline or rise of head normal to the surface, is _____.

Options :

1. Storativity
2. Transmissibility
3. Permeability
4. Porosity

Question Number : 52 Question Id : 1505611654 Question Type : MCQ Option Shuffling : No

The correlation between silt factor and mean particle diameter of silt in mm given as $f = 1.76\sqrt{m_r}$ by _____.

Options :

1. Lacey's theory
2. Kennedys theory
3. Kutters equation
4. Lindlen's theory

Question Number : 53 Question Id : 1505611655 Question Type : MCQ Option Shuffling : No

If wheat requires about 7.5 cm of water after every 28 days and the base period of wheat is 140 days, find out the value of delta for wheat.

Options :

1. 37.5 cm
2. 7.5 cm
3. 5 cm
4. 140 cm

Question Number : 54 Question Id : 1505611656 Question Type : MCQ Option Shuffling : No

The average annual rainfall over the whole India is estimated as _____.

Options :

1. 189 cm
2. 319 cm
3. 89 cm
4. 119 cm

Question Number : 55 Question Id : 1505611657 Question Type : MCQ Option Shuffling : No

Bernoulli's equation is derived making assumption that _____ .

Options :

1. The flow is steady, non-viscous incompressible and irrotational
2. The flow is uniform and incompressible
3. The flow is non viscous and steady
4. None of the above

Question Number : 56 Question Id : 1505611658 Question Type : MCQ Option Shuffling : No

Which of the following is a secondary air pollutant ?

Options :

1. Carbon monoxide
2. Sulphur dioxide
3. Ozone
4. Carbon dioxide

Question Number : 57 Question Id : 1505611659 Question Type : MCQ Option Shuffling : No

Air pollution causes harmful effects on _____ .

Options :

1. Biotic world
2. Abiotic world
3. Both (1) and (2)
4. None of the above

Question Number : 58 Question Id : 1505611660 Question Type : MCQ Option Shuffling : No

The first step in the treatment of sewage is to remove floating and suspended matter is _____ .

Options :

1. Screens
2. Grit chamber
3. Skimming tank
4. Grease traps

Question Number : 59 Question Id : 1505611661 Question Type : MCQ Option Shuffling : No

Which of the following treatment device is used for the treatment of aerobic biological suspended growth in waste water treatment ?

Options :

1. Screening
2. Grit removal
3. Activated sludge process
4. Chlorination

Question Number : 60 Question Id : 1505611662 Question Type : MCQ Option Shuffling : No

Reuse of treated sewage for urban landscaping primarily helps in _____ .

Options :

1. Increasing groundwater salinity
2. Reducing freshwater demand
3. Increasing treatment cost
4. Causing soil erosion

Question Number : 61 Question Id : 1505611663 Question Type : MCQ Option Shuffling : No

Which air pollution control device works primarily on the principle of inertial impaction ?

Options :

1. Electrostatic precipitator
2. Fabric filter
3. Cyclone separator
4. Wet scrubber

Question Number : 62 Question Id : 1505611664 Question Type : MCQ Option Shuffling : No

Which treatment method is most effective for removal of hardness in drinking water ?

Options :

1. Plain sedimentation
2. Chlorination
3. Hot lime-soda process
4. Rapid sand filtration

Question Number : 63 Question Id : 1505611665 Question Type : MCQ Option Shuffling : No

In the design of flocculation process, the term G means _____.

Options :

1. Gravitational force
2. Specific gravity of water
3. Velocity gradient
4. None of the above

Question Number : 64 Question Id : 1505611666 Question Type : MCQ Option Shuffling : No

The characteristic important for an adsorbent material is _____.

Options :

1. Porosity
2. Surface area to volume ratio
3. Specific gravity of material
4. None of the above

Question Number : 65 Question Id : 1505611667 Question Type : MCQ Option Shuffling : No

Indore method of composting is _____.

Options :

1. Anaerobic composting
2. Aerobic composting
3. Aerobic anaerobic both
4. Not specific

Increased amount of chlorides in river water indicates the discharge of _____.

Options :

1. Agricultural waste
2. Sewage effluent
3. Industrial discharge
4. Any other reason

Consider the following statements regarding design of a typical water supply scheme.

- I. Design period for conveying pipes is larger than that for service reservoirs.
- II. Design period for electric pumps is smaller than that for storage dams.

Options :

1. I and II are true
2. I and II are false
3. Only I is true
4. Only II is true

Why is the rate of filtration of gravity type rapid sand filter is larger than slow sand filters ? Select proper reason/s out of the following.

- a. Relatively large effective size of sand media.
- b. Relatively less uniformity coefficient of sand media.
- c. Applied water is pretreated with coagulation and settling.

Options :

1. Only a
2. Only b
3. Only c
4. All of a, b and c

Question Number : 69 Question Id : 1505611671 Question Type : MCQ Option Shuffling : No

How much is the suspended solids removal in primary waste water treatment, generally ? Select appropriate option of the following.

Options :

1. 5% to 10%
2. 20% to 30%
3. 55% to 70%
4. 95% to 99%

Question Number : 70 Question Id : 1505611672 Question Type : MCQ Option Shuffling : No

Which of the following is/are primary air pollutants ?

- a. Oxides of sulphur
- b. Oxides of carbon
- c. Ozone
- d. Sulphuric acid

Options :

1. Only a
2. Only b and c
3. Only a and b
4. Only a, c and d

Question Number : 71 Question Id : 1505611673 Question Type : MCQ Option Shuffling : No

The permissible gradient in the alignment of highway is called the _____.

Options :

1. Limiting gradient
2. Floating gradient
3. Ruling gradient
4. Exceptional gradient

The total extra widening required on a horizontal curve is given by _____ .

Options :

1.
$$W_e = \frac{nl^2}{2R} + \frac{V}{9.5R}$$

2.
$$W_e = \frac{nl^2}{2R} + \frac{V}{9.5\sqrt{R}}$$

3.
$$W_e = \frac{nl}{2R} + \frac{V}{9.5\sqrt{R}}$$

4.
$$W_e = \frac{nl^2}{2R} + \frac{V}{9.5R^2}$$

On permanent track, points and crossings are provided to change _____.

Options :

1. Gauge

2. Direction

3. Gradient

4. Space

The value of maximum super elevation prescribed on Indian railways for Broad Gauge track of group 'A' line is _____ .

Options :

1. 140 mm

2. 165 mm

3. 90 mm

4. 65 mm

The width of taxiway recommended by the ICAO are as for C type of airport _____.

Options :

1. 22.5 m
2. 15.0 m
3. 9.9 m
4. 7.5 m

The equation suggested by Horonjeff for the determination of turning radius of the taxiway is _____.

Options :

1.
$$R = \frac{0.388W^2}{\left(T - \frac{s}{2}\right)}$$
2.
$$R = \frac{0.388W^2}{\left(\frac{T}{2} - s\right)}$$
3.
$$R = \frac{0.388W^2}{\left(\frac{T}{2} + s\right)}$$
4.
$$R = \frac{0.388W^2}{\left(T + \frac{s}{2}\right)}$$

IRC has suggested the height of eyelevel of driver as 1.2 m and the height of the object as 0.15 m above the road surface, for the purpose of measuring _____.

Options :

1. Cross sight distance
2. Stopping sight distance
3. Overtaking sight distance
4. Lateral sight distance

Question Number : 78 Question Id : 1505611680 Question Type : MCQ Option Shuffling : No

For supersonic jet planes, the maximum turning radius is recommended as _____.

Options :

1. 150 m
2. 160 m
3. 170 m
4. 180 m

Question Number : 79 Question Id : 1505611681 Question Type : MCQ Option Shuffling : No

Indian Roads Congress (IRC) code providing guidelines for the design of flexible pavements is _____.

Options :

1. IRC 37
2. IRC 34
3. IRC 72
4. IRC 82

Question Number : 80 Question Id : 1505611682 Question Type : MCQ Option Shuffling : No

The slope of the line joining the crown and the edge of the road surface is called as _____.

Options :

1. Camber
2. Kerbs
3. Shoulders
4. None of the above

Question Number : 81 Question Id : 1505611683 Question Type : MCQ Option Shuffling : No

The traffic sign “HAIR PIN BEND LEFT” classified under _____.

Options :

1. Warning sign
2. Regulatory sign
3. Mandatory sign
4. Informatory sign

Consider the following statements in the context of traffic volume.

- A. Traffic volume is the actual number of vehicles observed or predicted to be passing a point during a given time interval.
- B. The rate of flow represents the number of vehicles passing a point during a time interval less than one hour, but expressed as an equivalent hourly rate.

Options :

- 1. Statement A is incorrect, but statement B is correct
- 2. Both statements are correct
- 3. Statement A is correct, but statement B is incorrect
- 4. Both the statements are incorrect

Consider the following statements.

- A. With increase in speed of stream of vehicles on a roadway, the average density per unit length increases.
- B. When the speed of the traffic flow decreases and becomes zero, the density attains maximum value.

Options :

- 1. Statement A is incorrect, but statement B is correct
- 2. Both statements are correct
- 3. Statement A is correct, but statement B is incorrect
- 4. Both the statements are incorrect

Width of carriage way recommended by the IRC for two lanes with raised Kerbs is _____ m.

Options :

- 1. 3.5
- 2. 5.5
- 3. 7.0
- 4. 7.5

The length of the vehicle and the length of the wheel base affects : _____.

- a. Extra width of pavement at horizontal curve
- b. Safe overtaking sight distance
- c. The minimum turning radius
- d. Width of traffic lane

Options :

- 1. a only
- 2. a and d only
- 3. a, b and c only
- 4. b, c and d only

The spatial resolution of a satellite image greatly influenced by _____.

Options :

- 1. Sensor's signal-to-noise ratio
- 2. Instantaneous Field Of View (IFOV)
- 3. Spectral bandwidth of the sensor
- 4. Atmospheric scattering

Triangulation surveying is based on the principle of _____.

Options :

- 1. Measurement of all sides
- 2. Measurement of all angles
- 3. One measured side and remaining angles
- 4. Direct distance measurement

Nadir point in the photogrammetry is also called as _____.

Options :

1. Camera point
2. Isocenter point
3. Photo plumb point
4. Focal point

The tangent length of simple circular curve is _____.

Options :

1. $T = R \tan \phi$
2. $T = R \tan \frac{\phi}{2}$
3. $T = \frac{\pi R \phi}{180^\circ}$
4. $T = R \sin \frac{\Delta}{2}$

Surveying is fundamentally concerned with _____.

Options :

1. Preparation of construction drawings
2. Measurement and representation of earth's surface features
3. Estimation of earthwork quantities
4. Design of engineering structures

Which of the following statement is/are correct ?

Statement A : In quadrantal bearing system the difference between corrected forebearing and corrected back bearing of same line is always exactly equals to $90^{\circ}00'$

Statement B : In whole circle bearing system the difference between corrected forebearing and corrected back bearing of same line is always exactly equals to $180^{\circ}00'$

Options :

1. Both statements are correct
2. Statement A is correct but statement B is incorrect
3. Statement A is incorrect, but statement B is correct
4. Both the statements are incorrect

If the scale is given as 1 cm = 1 km, then terms of RF it will be _____.

Options :

1. $\frac{1}{100}$
2. $\frac{1}{1,000}$
3. $\frac{1}{10,000}$
4. $\frac{1}{1,00,000}$

Salient features of a total station are _____.

- a. High accuracy
- b. Automatic atmospheric correction
- c. Long measuring range
- d. No error in observation

Options :

- 1. a and c only
- 2. a, c and d only
- 3. a, b and c only
- 4. a and d only

Which type of error can be minimized but not completely eliminated by careful observation ?

Options :

- 1. Gross errors
- 2. Systematic errors
- 3. Random errors
- 4. Instrumental errors

Which of the following statement is/are correct ?

Statement A : The reduced level of the plane of collimation is equal to the reduced level of the Benchmark plus the back sight reading taken on Benchmark.

Statement B : The reduced level of the plane of collimation is also equal to the reduced level of the change point plus back sight reading taken on change point.

Options :

1. Both statements are correct
2. Statement A is correct, but statement B is incorrect
3. Statement A is incorrect, but statement B is correct
4. Both statements are incorrect

The amount of refraction correction is _____.

Options :

1. $0.0785 D^2m$
2. $0.0112 D^2m$
3. $0.0673 D^2m$
4. $0.0587 D^2m$

In trigonometrical levelling, the vertical angle is measured with _____.

Options :

1. Compass
2. Theodolite
3. Total station
4. EDM

An aerial camera used in aerial photogrammetry is of _____.

Options :

1. Fixed focus type
2. Variable focus type
3. Auto focus type
4. Manual focus type

The major limitation of remote sensing data in engineering survey is _____.

Options :

1. Low spatial coverage
2. Dependence on ground truth
3. Requirement of skilled manpower
4. High cost of investments

Which GIS operation determines the spatial proximity ?

Options :

1. Overlay
2. Buffering
3. Spatial query
4. Geo-referencing